

Special Notice

Microsystems Technology Office (MTO) Spark Tank

DARPA-SN-25-40

March 14, 2025



Defense Advanced Research Projects Agency

Microsystems Technology Office

675 North Randolph Street

Arlington, VA 22203-2114

Special Notice DARPA-SN-25-40

Microsystems Technology Office (MTO) Spark Tank Defense Advanced Research Projects Agency (DARPA)

Posting Date: March 14, 2025

Event Date: July 24–25, 2025

Registration Opens: April 10, 2025

Registration Deadline: June 30, 2025, at 11:59 p.m. Eastern Time (ET)

Registration Website: <https://cvent.me/xAO84R>

Technical POC: Mr. Mike Geertsen, Deputy Director, MTO

E-mail: MTO-SparkTank@darpa.mil

DESCRIPTION:

The Defense Advanced Research Projects Agency (DARPA) will host the MTO Spark Tank event for prospective proposers on July 24–25, 2025, in Aurora, Colorado. **Advance registration is required. Registration will open on April 10, 2025. There will be no on-site registration. This event will be hosted in-person only.** The event will include overview presentations by DARPA personnel, including MTO Program Managers, and an opportunity for 1:1 engagement with MTO Program Managers and MTO Leadership via sidebar meetings. After the event, presented materials may be posted to <https://www.darpa.mil/research#research-opportunities>.

The goals of MTO Spark Tank are to: (1) familiarize participants with MTO's mission, interest areas, and on-going efforts; (2) facilitate discussions with potential proposers; (3) increase the efficiency in proposal preparation and evaluation; and (4) to broaden DARPA's reach with small businesses, educational institutions, and other non-traditional performers to break down barriers of entry for performers new to DARPA and the national security space.

This meeting will be held at the Unclassified level. All attendees are required to present government-issued photo identification upon entry to the event. The MTO Spark Tank event is open only to prospective MTO proposers and is closed to the general public and members of the media. **Federally Funded Research and Development Centers (FFRDC), University Affiliated Research Centers (UARC)s, and Government Entities are discouraged from attending this event.**

MTO MISSION AND THRUST AREAS:

MTO's mission is to dominate the microsystems ecology militarily and economically. To do this, MTO will pursue transformative microsystems science and technology research that result in 1) scientific disruption, by realizing photonic, quantum, and organic circuits that enable new differentiating capabilities; 2) ecosystem disruption, by establishing a new sustainable microsystem manufacturing ecosystem; and 3) market disruption, by creating asymmetric advantages by intelligent leveraging of commercial scaling with Department of Defense (DoD)-only capabilities through dual-use by design.

MTO's three thrusts are:

1. Developing technology to enable fundamentally new ways of creating useful circuits for the next generation of microsystems. Such technologies will not serve as replacement technologies, but rather opportunities to rethink microsystems with additional degrees of freedom. Specifically:
 - **Photonic circuits.** While photonics is not new to MTO, there is still much to explore through using the power of light at the chip-scale. MTO will expand the use of photonic integrated

circuits by examining new materials, processing methods, wavelengths, design architectures, and modeling and simulation tools.

- **Quantum circuits.** Quantum phenomena offer potentially dramatic improvements over classical computing and sensing. MTO will examine both nearer and longer-term instantiations of quantum technology to reduce the possibility of commercial surprise, drive the discovery of new hardware metrics, and realize scalable quantum devices.
 - **Bio/Organic circuits.** Biological and organic systems engage in complex computation and sensing activities every day at an efficiency and effectiveness that often dwarves its inorganic counterparts. MTO is looking to enable the integration of biomolecules and micro-technologies, establish the viability of molecular integrated circuits, and explore hybrid bio-sensing and bio-compute microsystems.
2. **Microsystems manufacturing ecosystem** is developing the infrastructure needed to ensure sustainable manufacturing capabilities for advanced microsystems. MTO is exploring new additive, subtractive, and hybrid fabrication tools as well as technologies to enable environmentally sustainable manufacturing, creating new tools to catalyze the manufacturing science needed to enable rapid progression from concept to product to support a first-to-market mentality, and foundational ecosystem enhancements with far-reaching impact beyond the circuits of tomorrow.
 3. **Dual use by design** is recognizing that the DoD must both take advantage of commercial scaling but also find ways to build superior capability into defense microsystems. MTO will explore new technologies in design, integration, and hardware security to create differentiating capabilities that leverage commercial scales to ensure both economic and national security in the microsystems domain.

REGISTRATION INFORMATION:

The MTO Spark Tank will be held in Aurora, Colorado on July 24–25, 2025 at the Gaylord Rockies Resort & Convention Center, located at 6700 North Gaylord Rockies Boulevard, Aurora, Colorado, 80019. The registration fee is \$450.00 for all attendees (standard and U.S. Military or U.S. Government Personnel). Attendance is open to individuals who are U.S. citizens, U.S. permanent residents, and foreign nationals. Registration is limited by the venue capacity (maximum of 550 participants); early registration is strongly recommended.

Registration will open on April 10, 2025. There will be no on-site registration. Advance registration is required.

Please register at <https://cvent.me/xAO84R> no later than June 30, 2025, at 11:59 p.m. ET. Late registrations may be admitted after all pre-registered attendees are admitted, if room capacity allows. Note that late registration processing requires significant coordination and may not be completed in time to attend the MTO Spark Tank event.

All meeting registrants who are not U.S. Citizens must complete and submit one of the following:

- All U.S. Permanent Residents and Foreign Nationals not representing a foreign government must submit a DARPA Form 60 “U.S. Permanent Resident and Foreign National Visit Request”, which can be submitted via the following link: <https://dtsn.darpa.mil/eform60/>.

- Foreign government personnel who are representing a foreign government in an official capacity must submit an official Foreign Visit Request (FVR) via their respective Embassy based in Washington, DC.

Failure to complete and submit the required forms prior to the registration deadline may result in the cancellation of existing registration. Please contact SIDInternationalSecurity@darpa.mil for any further questions about the security process.

CAPABILITY SUMMARY:

All attendees are required to submit a Capability Summary at the time of registration, by the registration deadline specified in this announcement. The Capability Summary (500-word text-only limit, no graphics/images) must follow the format provided within the registration site. This summary shall be stated in the attendee's own words and be a concise description of the core competencies of the entity (e.g., small business, research group, department, business unit, company, etc.) he or she represents. The summary should include research area(s) that are relevant to the interest(s) of the MTO Program Manager(s) with which sidebar requests are made. If applicable, describe accomplishments or ongoing work in this area within the previous five (5) years. **The Capability Summary should NOT propose new research topics, new project scoping, or solicitations of any kind for research funds from DARPA.** The Capability Summaries will be used to inform current and future MTO program managers of available competencies for appropriate follow-up and/or apprise Justification and Approval documents for other than Full and Open Competition solicitations if appropriate. Capability Summaries will not be distributed outside of DARPA.

MTO Spark Tank attendees are advised that DARPA will not acknowledge receipt of the information received or provide feedback to respondents with respect to any information submitted under this Special Notice.

No classified, ITAR controlled, or proprietary information shall be included in submitted materials or presented or discussed at the MTO Spark Tank event.

SIDEBAR MEETINGS:

Registered attendees will have the option to participate in a fifteen (15)-minute sidebar meeting with (1) MTO Program Managers for technical discussion, and or (2) MTO Leadership to learn about DARPA MTO Program Manager position opportunities. The attendees can request either one of the options or both options. For Option (1), attendees may request a sidebar meeting (via the registration website) with up to three of the MTO Program Managers listed below. For Option (2), attendees who are interested in a discussion related to the DARPA MTO Program Manager position may request a meeting (via the registration website) with MTO Leadership. The Capability Summary submission will assist in the coordination of the sidebar meetings; these meetings will be coordinated until capacity is met for each MTO person. Prior to the event, attendees selected for sidebar meetings will be notified of their meeting time and on-site instructions; **no additional sidebar meetings will be scheduled during the MTO Spark Tank event.**

To allow for the widest breadth of attendance within the in-person sidebar meetings, no more than two (2) representatives per organization unit (e.g., subsidiary, segment, sector, etc.), and eight (8) people per overall organization (e.g., parent company), may participate in the event. The determination of what constitutes an organization will be made by the DARPA MTO Office Director. Registrants are strongly encouraged to coordinate attendance internally within their organizations prior to registration and requesting to participate in the sidebar meeting(s).

MTO PROGRAM MANAGER INTERESTS:

1. [Dr. Justin Cohen](#) seeks to revolutionize optics with microsystems that do more than miniaturize free-space and fiber-optic systems, but rather provide unique functionality and performance attributes. Example microsystem topics of strong interest include:

- Integrated quantum photonics and nonlinear optics for sensing, especially exploiting non-classical states of light
- High-bandwidth quantum communication
- Ultra-low-noise microsystems leveraging chip-scale optics for sensing and communications
- Microsystems that create and exploit strong light-matter interactions and extreme optical nonlinearity
- Multiphysical microsystems exploiting couplings between optics, acoustics, radio frequency/microwave, etc.
- Photonic-electronic integration and co-design

Such systems likely require innovations in a combination of domains from materials, devices, circuits, integration and packaging, and applications that can effectively utilize the novel underlying capabilities. Concepts that focus solely on a single layer of this framework, such as Isolated demonstrations of individual photonic devices, and application demonstrations that primarily leverage existing photonic platforms, are of limited interest. As an application domain, signal processing and computing are also of limited interest.

2. [Dr. John M. Hoffman](#) is focused on reimagining the intersection of biology, computer science and microsystems. Topics of interest include: (1) integrating biomolecules and microsystems for sensing, actuation and computation applications, (2) molecular machines that perform important additive, subtractive, and hybrid materials fabrication, (3) synthetic biological systems that perform important scientific computing problems, and (4) the intersection of engineered biology and manufacturing to enable microsystem fabrication at scale.

Excluded topics include solutions that (1) do not break fundamental assumptions of biology's capability and utility, and (2) fail to address limitations that prevent scaling to useful problems.

3. [Dr. Jonathan Hoffman](#) seeks to rethink microsystems to create groundbreaking capabilities in sensing and communications. His interests include: (1) integrating quantum sensors into operational platforms for improved sensing, positioning, navigation, and timing in GPS-denied environments, (2) next-generation sensors supporting enhanced battlespace awareness and informed decision making, targeting and lethality, and increased platform survivability, (3) fielding non-traditional, resilient, and flexible communications that exceed traditional limitations, (4) noise-reduction methods to operate below the traditional noise floor, and (5) developing low-loss materials with controllable high permittivity and high permeability.

Excluded topics include evolutionary advancements in sensing, communications, and computing.

4. [Dr. Yogendra Joshi](#) seeks revolutionary advancements in microsystem capabilities through improved energy management. Focus areas of interest include: (1) high-efficiency conversion of low-grade waste heat into electricity, (2) energy efficient cryogenic operation from 77 K to mK, (3) advances in surface thermal radiation for dynamic energy management of terrestrial and space-based microsystems.

Additional areas of interest include: (1) energy management in heterogeneously integrated microsystems, (2) machine learning accelerated multi-scale, multi-physics computations, and (3) compact cryogenic refrigeration.

Evolutionary advances in existing energy management technologies are of limited interest.

5. [Dr. Julian McMorro](#) has varied technical interests, spanning microsystem manufacture and energy generation/storage in supply-constrained environments; e-waste elimination; large-scale assembly of nanomaterials; living microsystems for computing, sensing, and control; and radiation-hardened electronics technologies as key mission enablers. Materials-driven solutions that leverage emergent behavior or assembly techniques are of particular interest.

Excluded topics with limited interest include incremental improvements on existing technologies and approaches with limited prospects for scaleup.

6. [Dr. David Meyer](#) seeks to realize new microsystems that leverage emerging materials and manufacturing methods to enable broad-impact, leap-ahead capabilities for national security. He is interested in a range of topics including (1) electronic, optoelectronic, and acoustic applications of ultrawide bandgap materials and devices, (2) parallelization of microscale and nanoscale additive manufacturing, and (3) multifunctional distributed membrane microsystems.

Additional areas of interest include (1) developing technology to assist in supply chain security of rare earth and precious metals, (2) AI-assisted radio-frequency integrated chip (RFIC) design, and (3) increasing the throughput of materials growth systems by 10x.

Excluded topics of limited interest include (1) electronic devices based on 2D materials, (2) evolutionary device development where performance metric is not improved by 10x or more, and (3) simple geometric scaling of existent technologies to application-relevant sizes or volumes.

7. [Mr. Daniel Ridge](#) seeks to combine high-performance computing (HPC), vulnerability assessment, media forensics, and analyses of the design and lifecycle of embedded systems into novel approaches to outstanding problems in cyber security.

Mr. Ridge is not interested in techniques that use HPC to scale traditional forensic and vulnerability analysis techniques without also proposing the expansion of these techniques to new classes of vulnerabilities, media, or embedded devices.

8. [Mr. Michael Sangillo](#) seeks to rethink microsystem constructs to create groundbreaking capabilities in exploitation missions. He is focused on overcoming technical challenges in (1) sensor systems, (2) microsystem manufacturing, and (3) cyber-physical systems. He is inspired by the idea of domain convergence where high-impact opportunities are realized by merging multiple technical domains.

Additional areas of interest include: (1) developing techniques that mitigate artificial intelligence-informed threats, (2) autonomous exploitation, and (3) technologies that enable rapid de-escalation of conflict. Excluded topics with limited interest include: (1) evolutionary semiconductor manufacturing and (2) better mousetrap fallacies.

9. [Dr. Thomas Schratwieser](#) is focused on novel power storage and generation to enable DoD missions where size, weight, and power (SWaP) and cost are significant factors. He is also interested in

(1) applications for high-energy EM sources, (2) methods for passive, GPS-free navigation, and (3) long-distance detection of nuclear materials.

Additional topics of interest include (1) uses for exquisitely synchronized moving platforms, (2) tunable metasurfaces for lenses, and (3) accurate, high-bandwidth accelerometry.

Excluded topics include new atomic clocks and new magnetometers.

10. [Dr. Anna Tauke-Pedretti](#) seeks to increase access to nontraditional semiconductor materials and photonic circuit technology to enable new capabilities in sensing, compute and quantum systems. She is focused on overcoming technical challenges associated with 1) low-volume microsystem manufacturing, 2) scalable photonic integrated circuits, and 3) the heterogeneous integration of semiconductors.

Excluded topics with limited interest include incremental improvements in photonic device technology.

11. [Ms. Trish Veeder](#) is exploring what DoD missions become possible with an adaptive interface between camera hardware and artificial intelligence (AI) / machine learning (ML). Algorithms that enable new capabilities and novel camera hardware needed to support end to end training with these algorithms are especially interesting. Additional areas of interest include (1) electro-optic and infrared countermeasures, (2) high performance tunable optics, and (3) supporting manufacturing ecosystems.

Topics of limited interest include (1) infrared absorbers, (2) commercial AI/ML, and (3) evolutionary improvements to current methods.

12. [Dr. Huanan Zhang](#) is passionate about tackling complex technical challenges at the intersection of micro/biosystems. He is focused on (1) advanced semiconductor processing, (2) microsystems for critical medical care, (3) novel bioelectronic applications, and (4) organic computing platforms. He believes in the power of interdisciplinary collaboration, leveraging expertise from various scientific and engineering fields to generate high-impact solutions for mission-critical applications.

Excluded topics with limited interest include: (1) evolutionary semiconductor manufacturing, and (2) conventional medical devices.

TEAMING:

To assist those wanting to establish collaborative teaming efforts and business relationships, potential proposers may indicate on the registration website that they authorize their contact information to be shared among meeting participants. For organizations new to DARPA, this may be of use in connecting with potential partners. Team formation, communications, and networking are the sole responsibility of the participants.

CONTACT:

- Registration Website: <https://cvent.me/xAO84R>
- Administrative Questions: MTO-SparkTank@darpa.mil
- Technical Questions: MTO-SparkTank@darpa.mil
- Security Process: SIDInternationalSecurity@darpa.mil

Please refer to the MTO Spark Tank Special Notice number (DARPA-SN-25-40) in all correspondence.

DARPACONNECT:

Entities who have not worked with DARPA before are encouraged to learn more about DARPAConnect, an initiative established to facilitate collaboration between DARPA and potential performers. The DARPAConnect team offers customized support, resources, and guidance on how to prepare your ideas for high-impact conversations with DARPA program managers. Please visit [DARPAConnect.us](https://darpaconnect.us) to access a digital hub of online resources, including a curriculum for self-paced learning, personalized support, and in-person and virtual events. In addition to the self-paced online materials, the DARPAConnect team is able to schedule one-on-one conversations to discuss your specific ideas, questions, and paths to DARPA. You can use the contact form at [DARPAConnect.us](https://darpaconnect.us) or email the DARPAConnect team directly at darpaconnect@darpa.mil to request assistance.

DISCLAIMERS:

This Special Notice is issued solely for information and program planning purposes and does not constitute a formal solicitation for proposals or proposal abstracts; any so sent will be disregarded. In accordance with FAR 15.201(e), responses to this Special Notice are not offers and cannot be accepted by the Government to form a binding contract. Attendance is strictly voluntary and is not required to propose to subsequent Announcements (if any) or Solicitations (if any) on this topic. DARPA will not provide reimbursement for costs incurred in responding to this Special Notice. Respondents are advised that DARPA is under no obligation to acknowledge receipt of any information received or provide feedback to respondents with respect to any information submitted under this Special Notice.

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